

The Dark Unification

Self-Interacting Majorana Dark Matter, Path-Integral Foundations,
and Dark Energy as the T-Breaking Component

Omer P.

Independent Researcher

github.com/0mp3s

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Abstract

We present a unified theoretical framework for the dark sector built from a single Lagrangian with three free parameters (m_χ, m_ϕ, α) . **Chapter 1** establishes the phenomenological model: a Majorana fermion dark matter candidate χ interacting via a real scalar mediator ϕ . Velocity-dependent self-interaction cross sections are computed exactly via the Variable Phase Method (VPM), and an MCMC scan over astrophysical data identifies 122 viable benchmark points consistent with relic density, Fornax globular cluster survival, SPARC rotation curves, the Radial Acceleration Relation, and direct/indirect detection constraints. **Chapter 2** derives all Chapter 1 results from the generating functional $Z[J]$ of the path integral, proving that VPM is *exactly* the fluctuation determinant of the scattering field integral (no approximation). The Coleman-Weinberg potential, finite- T Matsubara formalism, and dark QCD misalignment together predict $H_0 = 67.4$ km/s/Mpc from $V_{eff}(\sigma_0) = \rho_\Lambda$ via the Friedmann equation. **Chapter 3** proposes a Dark Electromagnetic Duality: the CP phase $\theta \equiv \sigma/f$ is promoted to a dynamical axion field, with $y_s = y \cos(\sigma/f)$ (T-even, SIDM clustering) playing the role of $\vec{E}$ and $y_p = y \sin(\sigma/f)$ (T-odd, vacuum energy) playing the role of $\vec{B}$. Dark energy is not a separate phenomenon; it is the T-violating projection of the same Yukawa interaction. The universal relic angle $\theta_{\text{relic}} = \arctan(1/\sqrt{8}) = 19.47^\circ$ is fixed simultaneously by SIDM and relic density constraints, with no free parameters.

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Notation and Conventions

- Metric signature: $(+, -, -, -)$
- Natural units: $\hbar = c = k_B = 1$ unless stated
- $\alpha = y^2/(4\pi)$, $\alpha_s = y_s^2/(4\pi)$, $\alpha_p = y_p^2/(4\pi)$
- $\lambda = \alpha m_\chi/m_\phi$ (dimensionless coupling)
- $\kappa = \mu v_{\text{rel}}/m_\phi$ (dimensionless momentum), $\mu = m_\chi/2$
- Lagrangian sign convention: $\mathcal{L} = T - V$, so $V(r) < 0$ for attraction

Part I

Secluded Majorana SIDM: Phenomenology

1 Introduction: The Problem with Small Galaxies

On cosmological scales, the Λ CDM model works spectacularly well. But zoom in to individual dwarf galaxies and something breaks: the predicted dark matter densities in galactic cores are far higher than observed (the “core-cusp problem”); too many small satellite galaxies are predicted but never seen (“missing satellites”); and the largest predicted dwarfs are too dense to match any observed system (“too-big-to-fail”).

All three problems share a common fix: if dark matter particles scatter off each other, they redistribute energy from the dense inner halo to the outer region, naturally flattening the cusp into a core. This is self-interacting dark matter (SIDM).

The question is: what mediates this self-interaction, and does it remain consistent with everything else we know? This paper answers that question with a single, minimal model.

2 The Model: One Coupling, Three Parameters

We consider a Majorana fermion χ (the dark matter candidate) interacting with a real scalar mediator ϕ through a Yukawa interaction. The choice of Majorana, rather than Dirac, is not arbitrary: a Majorana fermion is its own antiparticle, which halves the relic density calculation and introduces non-trivial quantum statistics in the self-interaction (see below). The mediator ϕ has *no coupling to the Standard Model* — the dark sector is secluded. This single choice eliminates constraints from direct detection and CMB energy injection simultaneously.

The interaction is simply:

$$\mathcal{L}_{\text{int}} = -\frac{y}{2}\bar{\chi}\chi\phi \quad (1)$$

In the non-relativistic limit relevant to galactic dynamics, this generates an attractive Yukawa potential between two DM particles:

$$V(r) = -\frac{\alpha}{r}e^{-m_\phi r}, \quad \alpha = \frac{y^2}{4\pi} \quad (2)$$

The range of the force is set by $1/m_\phi$; the strength by α . Three numbers — (m_χ, m_ϕ, α) — describe the entire model.

2.1 Scalar Potential and Vacuum Stability

For a stable relic ϕ abundance, a cubic self-coupling μ_3 enables $3\phi \leftrightarrow 2\phi$ “cannibal” number-changing processes that prevent ϕ from overclosing the universe. This requires $\mu_3/m_\phi \gtrsim 1.7$ [1]. However, a non-zero μ_3 tilts the scalar potential and can create a second vacuum deeper than $\phi = 0$, destabilizing the theory.

The full potential is:

$$V(\phi) = \frac{1}{2}m_\phi^2\phi^2 + \frac{\mu_3}{3!}\phi^3 + \frac{\lambda_\phi}{4!}\phi^4 \quad (3)$$

Demanding that $\phi = 0$ remain the global minimum (no deeper secondary vacuum) yields the stability condition:

$$\lambda_\phi > \frac{3\mu_3^2}{8m_\phi^2} \quad (4)$$

At the cannibal threshold $\mu_3/m_\phi = 1.7$, this gives $\lambda_\phi \gtrsim 1.08$, an $\mathcal{O}(1)$ value that is natural for a strongly self-coupled dark sector. A numerical scan confirms the tighter bound $\lambda_\phi \gtrsim 0.96$. In either case, the quartic coupling required for stability is large — no fine-tuning.

3 Self-Interaction Cross Section: Why a Simple ODE is Exact

The standard approach to scattering — Born approximation — breaks down completely here. At $\lambda = \alpha m_\chi/m_\phi \sim 2\text{--}49$ (our parameter space), Born overestimates the true cross section by factors of 88–135. A full partial-wave calculation is unavoidable.

The Variable Phase Method (VPM) solves directly for the phase shifts δ_ℓ of each partial wave via a first-order ODE:

$$\frac{d\delta_\ell}{dx} = +\frac{\lambda e^{-x}}{\kappa x} \left[\hat{j}_\ell(\kappa x) \cos \delta_\ell - \hat{n}_\ell(\kappa x) \sin \delta_\ell \right]^2 \quad (5)$$

where $x = m_\phi r$ (dimensionless radius), $\kappa = \mu v_{\text{rel}}/m_\phi$ (dimensionless momentum), and $\lambda = \alpha m_\chi/m_\phi$ (coupling strength). Integrating from $x = 0$ to $x \gg 1$ gives the asymptotic phase shift directly.

For identical Majorana fermions, there is a subtlety: since $\chi = \bar{\chi}$, the two-body wavefunction must be antisymmetric under particle exchange. This forces even- ℓ partial waves into the spin-singlet channel (weight 1) and odd- ℓ into the spin-triplet (weight 3), giving:

$$\sigma_T = \frac{2\pi}{k^2} \left[\sum_{l \text{ even}} (2l+1) \sin^2 \delta_l + 3 \sum_{l \text{ odd}} (2l+1) \sin^2 \delta_l \right] \quad (6)$$

The factor 2π rather than 4π accounts for identical-particle kinematics.

Why is this exact, not approximate? Chapter 2 proves that this ODE is precisely the fluctuation determinant of the path integral over ϕ in the two-body background — no perturbative expansion anywhere (see Appendix F).

4 Relic Density: The Same Coupling Does Both Jobs

Dark matter freezes out when the annihilation rate $\Gamma = n\langle\sigma v\rangle$ falls below the Hubble expansion rate. For Majorana $\chi\chi \rightarrow \phi\phi$ (t- and u-channel scalar exchange), the leading term is s-wave:

$$\langle\sigma v\rangle_0 = \frac{\pi\alpha^2}{4m_\chi^2} \quad (7)$$

This is the crucial feature of the model: the *same* coupling α that controls the SIDM cross section at galactic velocities ($v \sim 30\text{--}1000$ km/s) also determines the relic abundance at freeze-out ($v \sim 0.3c$). There is no freedom to tune one without the other.

The Sommerfeld enhancement at freeze-out is negligible: $S_0 < 1.026$ for all viable benchmarks, a correction under 2.5%. The secluded dark sector means annihilation products never reach the SM plasma, so CMB energy-injection constraints do not apply.

5 Numerical Pipeline and Validation

Before presenting results, we describe the computational pipeline and its validation, since every claim in this paper rests on the numerical cross section solver.

5.1 VPM Solver Implementation

The VPM ODE (Eq. 6) is integrated using a 4th-order Runge-Kutta scheme with adaptive step size, from $x = 10^{-4}$ to $x_{\max} = 50\text{--}200$ (chosen to ensure $e^{-x_{\max}} < 10^{-10}$). Partial waves are summed to $\ell_{\max}$ determined by a convergence criterion:

$$\ell_{\max} = \text{smallest } \ell \text{ such that } \left| \frac{\delta_\ell}{\delta_0} \right| < 10^{-6} \text{ for three consecutive } \ell \quad (8)$$

In practice, $\ell_{\max} = 3\text{--}8$ for BP1 ($\lambda = 1.9$) and $\ell_{\max} = 30\text{--}80$ for MAP ($\lambda = 48.6$). The solver is JIT-compiled via `numba` for performance.

5.2 Cross-Check Suite

We perform six independent validation tests on the VPM solver:

1. **Born limit.** At $\lambda \ll 1$, VPM must reproduce the analytic Born cross section. Over 5 test points with $\lambda = 0.05\text{--}0.75$: $\sigma_T^{\text{VPM}}/\sigma_T^{\text{Born}} = 1.000 \pm 0.002$.
2. **Unitarity.** Every phase shift must satisfy $|\sin^2 \delta_\ell| \leq 1$. Tested on all 122 viable benchmarks at 9 velocities ($122 \times 9 = 1098$ evaluations): zero failures, worst violation 4.3% below the bound.
3. **Transfer $\equiv$ elastic.** For identical Majorana fermions, the momentum-transfer cross section $\sigma_T^{\text{transfer}}$ equals the elastic cross section $\sigma_T^{\text{elastic}}$ exactly, because within each spin channel the symmetrized scattering amplitude satisfies $|f(\theta)|^2 = |f(\pi - \theta)|^2$, making $\int \cos \theta d\sigma = 0$. We verify this identity numerically to $|\sigma_T^{\text{tr}}/\sigma_T^{\text{el}} - 1| < 10^{-12}$ at all velocities for both BP1 and MAP. This is a non-trivial check of the Majorana spin-statistics implementation.
4. **Literature comparison.** Validation against the Tulin, Yu & Zurek [2] classical-regime formula (their Eq. 14): agreement within $\mathcal{O}(1)$ factors in the resonant regime, and exact agreement in the Born regime, as expected.
5. **Blind off-grid stress test.** 200 randomly generated parameter points never used in fitting: 100 within the viable bounds, 50 across the full space, 50 perturbations of known viable points. All 200 return physical (finite, positive) cross sections. No solver crashes or NaN failures.

6. **Preprint consistency.** BP1 cross sections reproduced identically from the published code: $\sigma_T/m(30) = 0.516$, $\sigma_T/m(1000) = 0.072 \text{ cm}^2/\text{g}$ (0% deviation).

5.3 Born Approximation Failure

In our parameter space ($\lambda = 0.7\text{--}49$), the Born approximation overestimates the true cross section dramatically. Table 1 shows the ratio $R_{\text{VB}} = \sigma_T^{\text{VPM}}/\sigma_T^{\text{Born}}$ for representative benchmarks:

Table 1: VPM/Born ratio R_{VB} at selected velocities. Born overestimates by factors of $2\text{--}5\times$ at dwarf velocities, converging only at cluster scales ($v \gtrsim 1000 \text{ km/s}$).

$v \text{ [km/s]}$	BP1 ($\lambda = 1.9$)			MAP ($\lambda = 48.6$)		
	VPM	Born	R_{VB}	VPM	Born	R_{VB}
12	0.430	1.87	0.23	1.22	4.84	0.25
30	0.516	1.81	0.29	1.71	5.85	0.29
200	0.356	0.742	0.48	—	—	—
1000	0.072	0.085	0.85	0.203	0.215	0.95
4700	0.004	0.004	0.92	—	—	—

This failure is not surprising: the Born approximation is a first-order perturbative expansion in λ , and breaks down at $\lambda \gtrsim 1$. What is noteworthy is the *magnitude* — factors of $3\text{--}5\times$ at the velocities most relevant to the small-scale problems. Any analysis of this model that relies on Born-level cross sections is unreliable.

5.4 Boltzmann Solver Accuracy

The relic density is computed both analytically (Kolb-Turner freeze-out approximation) and numerically (stiff ODE solver for the Boltzmann equation $dY/dx = -\lambda_{\text{ann}}[Y^2 - Y_{\text{eq}}^2]$). The numerical solver corrects the Kolb-Turner coupling by $\Delta\alpha = 16\%$ and the resulting cross section by $\Delta(\sigma_T/m) = 19\%$ for BP1 — a correction that exceeds many observational uncertainties and justifies the full numerical treatment.

6 MCMC Analysis and Parameter Posteriors

With three free parameters and multiple observational constraints, the critical question is whether any parameter space survives at all.

6.1 Setup

A Bayesian MCMC scan (`emcee` [3], 32 walkers, 5000 production steps after 500 burn-in, JIT-accelerated VPM solver) samples the posterior over (m_χ, m_ϕ, α) with flat priors: $\log_{10}(m_\chi/\text{GeV}) \in [0.5, 2.0]$, $\log_{10}(m_\phi/\text{MeV}) \in [0.5, 1.5]$, $\log_{10} \alpha \in [-4.0, -2.0]$.

The likelihood includes 13 astrophysical observables spanning four decades in velocity: dwarf galaxy velocity dispersions ($v \sim 12\text{--}60 \text{ km/s}$), group-scale ellipticals ($v \sim 250\text{--}280 \text{ km/s}$), cluster mergers ($v \sim 1000\text{--}1200 \text{ km/s}$), and the Bullet Cluster ($v \sim 4700 \text{ km/s}$), following Kaplinghat, Tulin & Yu [4]. Upper limits (Bullet Cluster, Harvey et al.) are treated as one-sided constraints: $\chi^2 = 0$ when $\sigma_T/m < \sigma_{\text{obs}}$.

6.2 Convergence Diagnostics

- Acceptance fraction: 0.542 (within optimal range 0.2–0.7).
- Maximum autocorrelation time: $\tau_{\max} = 75.1$ steps.
- Effective independent samples: $N_{\text{eff}} \approx 2132$.
- Gelman-Rubin diagnostic: $\hat{R} = (1.004, 1.005, 1.005)$ for (m_χ, m_ϕ, α) — well below the $\hat{R} < 1.1$ threshold.

6.3 Posteriors

The parameter posteriors are unimodal with the following 68% credible intervals:

$$m_\chi = 35.7^{+52.4}_{-25.6} \text{ GeV} \quad (9)$$

$$m_\phi = 8.4^{+4.3}_{-3.4} \text{ MeV} \quad (10)$$

$$\alpha = 10^{-3.0 \pm 0.6} \quad (11)$$

A connected island of viability survives:

$$m_\chi \in [10, 100] \text{ GeV}, \quad m_\phi \in [7.6, 14.8] \text{ MeV}, \quad \alpha \in [5 \times 10^{-4}, 5 \times 10^{-3}] \quad (12)$$

122 benchmark points satisfy all constraints simultaneously (80,142 raw viable, 646 representative after thinning, 394 BBN-safe). The island is narrow in m_ϕ (width ~ 7 MeV), setting the mediator mass at a scale where the Yukawa range $1/m_\phi \sim 25$ fm is comparable to the de Broglie wavelength at dwarf velocities — the resonant regime.

7 Confronting the Universe: Detailed Results

We test the model against every available astrophysical constraint, spanning five decades in velocity from ultra-faint dwarfs ($v \sim 12$ km/s) to the Bullet Cluster ($v \sim 4700$ km/s).

7.1 χ^2 Goodness of Fit

The velocity-dependent cross section is compared against 13 observational targets from Kaplinghat et al. [4], supplemented with Fornax globular cluster constraints from Read et al. [5].

The unconstrained best-fit achieves:

$$\chi^2_{\min}/\nu = 1.22/10 = 0.12 \quad (m_\chi = 100 \text{ GeV}, m_\phi = 9.2 \text{ MeV}, \alpha = 2.86 \times 10^{-3}) \quad (13)$$

Under the relic density constraint ($\Omega h^2 = 0.120 \pm 0.001$), the best-fit is:

$$\chi^2_{\text{relic}}/\nu = 3.82/10 = 0.38 \quad (14)$$

All 17 relic-viable benchmark points achieve $\chi^2/\nu < 1$, with the full ranking shown in Table 2.

Table 2: Best 5 relic-constrained benchmark points ranked by χ^2/ν . All 17 achieve $\chi^2/\nu < 1$; the full table is in Appendix I.

Rank	m_χ [GeV]	m_ϕ [MeV]	α	λ	χ^2/ν
1	20.69	9.91	1.05×10^{-3}	2.1	0.38
2	29.76	11.34	1.47×10^{-3}	3.9	0.41
3	14.38	8.66	7.56×10^{-4}	1.3	0.42
4	37.93	12.98	1.87×10^{-3}	5.5	0.44
5	42.81	13.15	2.09×10^{-3}	6.8	0.46

Table 3: Predicted σ_T/m [cm²/g] for BP1 and MAP benchmarks compared with observational targets. Upper limits are one-sided.

System	v [km/s]	BP1	MAP	Observed
Draco	12	0.430	1.22	(unconstrained)
TBTF dwarfs	30	0.516	1.71	1–3
NGC 2976	60	0.499	—	~ 0.5
NGC 720	250	0.325	—	0.1–1
Abell 3827	1000	0.072	0.203	< 1.5
Abell 2537	1100	0.061	—	< 1.0
Abell 370	1200	0.052	—	< 1.0
Bullet Cluster	4700	0.004	—	< 1.25

7.2 Velocity-Dependent Cross Sections

Table 3 shows the predicted σ_T/m at 10 characteristic velocities alongside observational constraints. The key feature is the velocity dependence: strong self-interaction at dwarf scales ($\sigma_T/m \sim 0.5$ – 1.7 cm²/g at $v \sim 30$ km/s) and natural suppression at cluster scales ($\sigma_T/m \sim 0.07$ – 0.2 cm²/g at $v \sim 1000$ km/s).

7.3 Fornax Globular Cluster Survival

The Fornax dSph contains five globular clusters (GC1–GC5) that orbit within ~ 1 kpc of the center. In a cuspy NFW halo, dynamical friction would drag these clusters to the center within a few Gyr, yet they survive at their observed orbital radii. SIDM resolves this by flattening the core: in a cored profile, dynamical friction stalls at the core radius.

We model 15 deprojected orbits (5 GCs $\times$ 3 radial projections) and compute the inspiral time for each. For MAP:

- 14 out of 15 deprojections stall inside the SIDM core ($r_{\text{core}} = 829$ pc), matching the observed $r_{\text{core}} = 500$ – 900 pc.
- Jeans equation fit with Osipkov-Merritt anisotropy: $\chi^2 = 2.3$. For comparison, an NFW profile gives $\chi^2 = 152$.

Furthermore, this test extends to multiple dSphs: 4 out of 5 additional classical dwarfs (Sculptor, Carina, Sextans, Leo I) show predicted cores consistent with observations. This is not a single object — it is a population-level test.

7.4 Radial Acceleration Relation

The McGaugh et al. [6] Radial Acceleration Relation (RAR) in SPARC spiral galaxies shows a tight correlation between observed and baryonic accelerations. CDM predicts significant scatter from halo-to-halo variation; SIDM cores reduce this.

For BP1, the RAR scatter decreases from 0.177 dex (observed/CDM) to 0.122 dex, a 31% reduction. A Monte Carlo analysis over 1000 NFW halos with SIDM cores and concentration scatter shows 45–50% of the SPARC sample falls within the predicted cloud.

7.5 Secluded Sector: Direct Detection and Cosmological Safety

The secluded dark sector (ϕ has no tree-level SM coupling) has two critical consequences:

Direct detection. The spin-independent cross section $\sigma_{\text{SI}} = 0$ at tree level. A Higgs portal coupling $\lambda_{h\phi}|H|^2\phi^2$ is generically excluded: the BBN constraint on ϕ lifetime requires $\sin\theta_{\text{mix}} \gtrsim 2 \times 10^{-5}$, but the LZ experiment constrains $\sin\theta_{\text{mix}} \lesssim 6 \times 10^{-10}$ for $m_\phi \sim 10$ MeV. The gap is a factor of 3×10^4 — *the portal is generically closed*.

ΔN_{eff} . At BBN ($T \sim 1$ MeV), $m_\phi/T_\phi \sim 22$, giving $\Delta N_{\text{eff}} \sim e^{-22} \approx 3 \times 10^{-10}$. At CMB decoupling, $m_\phi/T_\phi \sim 10^8$. The maximum across the viable region is $\Delta N_{\text{eff}} \sim 7 \times 10^{-12}$, with a safety margin exceeding 10^{10} relative to the Planck bound $\Delta N_{\text{eff}} < 0.30$.

Mediator relic abundance. $\Omega_\phi/\Omega_\chi \sim 5 \times 10^{-4}$ — negligible. The cannibal mechanism ($3\phi \leftrightarrow 2\phi$) efficiently depletes the ϕ population.

7.6 Sommerfeld Enhancement at Freeze-out

The Sommerfeld factor S_0 at freeze-out ($v \sim 0.3c$, $\kappa = 50\text{--}760$) ranges from 1.003 to 1.025 across all 17 relic benchmarks — a correction of at most 2.5%. The s-wave tree-level calculation is therefore valid. At lower velocities ($v \sim 30$ km/s), Sommerfeld enhancement reaches $S_0 \sim 5\text{--}400$, but since the dark sector is secluded, late-time annihilation products do not inject energy into the SM plasma, and CMB s-wave constraints do not apply.

7.7 Falsifiable Predictions

The model makes several sharp, testable predictions:

1. **Majorana fingerprint at $v \sim 40\text{--}95$ km/s.** For MAP ($\lambda = 48.6$), the ratio $\sigma_T^{\text{Maj}}/\sigma_T^{\text{Dir}}$ exceeds unity in a narrow velocity window around $v \approx 57$ km/s. This $R > 1$ “bump” is a direct consequence of odd- ℓ partial waves entering the spin-triplet channel (weight 3) and cannot occur in any Dirac+scalar model. Future IFU spectroscopy of intermediate dSphs (e.g., Leo II, $v_{\text{max}} \approx 50$ km/s) could probe this regime.
2. **No supermassive BH seeds.** Gravothermal core collapse could in principle form SMBH seeds. For all viable benchmarks, the gravothermal timescale exceeds the Hubble time by a factor $> 10^4$: $t_{\text{gc}}/t_{\text{univ}} > 10^4$. This is a *negative* prediction — the model does not form early SMBH seeds.
3. **Read+2019 margin.** The MAP point passes the Read et al. [5] low-velocity constraint ($\sigma_T/m < 1.5$ cm²/g at $v \sim 12$ km/s) with a margin of only $+0.05$ cm²/g. A $\sim 3\%$ tighter bound would exclude MAP.

4. **Ultra-faint dwarf cores.** MAP predicts detectable cores in 2 out of 6 ultra-faint dwarfs at current sensitivity, rising to observable in future surveys (LSST, Nancy Grace Roman). Classical dSphs: 7 out of 8 should show cores by $t = 12$ Gyr.

Part II

Path-Integral Foundations: From $\mathcal{L}$ to H_0

8 The Central Question

Chapter 1 computed scattering cross sections by solving a first-order ODE for phase shifts. It computed relic density by solving a Boltzmann equation. Both are standard tools — but where do they come from, really?

This chapter shows that both emerge as exact evaluations of the path integral over the Lagrangian of Chapter 1. No approximation is introduced at any step. The payoff goes beyond rigor: the same path integral, evaluated at one loop, a) proves the ODE is exact (not a numerical method), and b) predicts $H_0 = 67.4$ km/s/Mpc from dark QCD confinement.

The derivation proceeds in seven layers, each answering a specific physical question:

Table 4: Seven layers from Lagrangian to observable.

Layer	Question	Method	Answer
1	What carries the force?	Gaussian integration	Propagators Δ_F, S_F
2	What is the potential?	Tree-level (t+u)	$V(r) = -\alpha e^{-m_\phi r}/r$
3	How large is σ_T ?	Fluctuation determinant	VPM $\equiv$ exact PI
4	Does the vacuum prefer CP violation?	One-loop CW	A_4 vacuum at $\theta \neq 0$
5	Is that vacuum stable at freeze-out?	Matsubara / finite- T	Yes, CW dominates
6	Can the vacuum tunnel away?	Instanton	No ($S_E \sim 10^{121}$)
7	What is Ωh^2 ?	$\langle \sigma v \rangle + \text{Boltzmann}$	0.120
DE	What is H_0 ?	Dark QCD misalignment	67.4 km/s/Mpc

9 Layer 1: What Carries the Force?

We start from the full interacting path integral:

$$Z[J, \eta, \bar{\eta}] = \int \mathcal{D}\phi \mathcal{D}\chi \mathcal{D}\bar{\chi} \exp\left(i \int d^4x \mathcal{L} + J\phi + \bar{\eta}\chi + \bar{\chi}\eta\right) \quad (15)$$

Since all fields enter quadratically in the free Lagrangian, the Gaussian integrations

can be performed exactly. The result is two propagators:

$$\Delta_F(q^2) = \frac{i}{q^2 - m_\phi^2 + i\varepsilon} \quad (\text{scalar: spin-0, real}) \quad (16)$$

$$S_F(p) = \frac{i(\not{p} + m_\chi)}{p^2 - m_\chi^2 + i\varepsilon} \quad (\text{Majorana fermion}) \quad (17)$$

Physically, Δ_F tells us the force carrier propagates a distance $r_0 = 1/m_\phi$ before its amplitude falls by e^{-1} . This is the range of the force. For $m_\phi \sim 11$ MeV, $r_0 \approx 18$ fm — a scale where the de Broglie wavelength of DM at dwarf velocities ($\lambda_{dB} \sim 4/(m_\chi v) \approx 145$ fm for BP1) is comparable. This near-resonant condition is why the viable island exists at these particular masses.

10 Layer 2: Tree-Level Yukawa Potential

At tree level, two DM particles exchange a single ϕ through t- and u-channel diagrams. Evaluating the amplitude at small momentum transfer $|\mathbf{q}|^2 \ll m_\phi^2$ and Fourier transforming:

$$i\mathcal{M} = \left(-\frac{iy}{2}\right)^2 \frac{i}{-|\mathbf{q}|^2 - m_\phi^2} \longrightarrow V(r) = -\frac{\alpha}{r} e^{-m_\phi r} \quad (18)$$

This is the same Yukawa potential assumed in Chapter 1 — now derived, not postulated.

The Born approximation for the cross section is valid only when $\lambda \ll 1$. At our benchmark couplings ($\lambda \sim 2\text{--}49$), Born overestimates the true cross section by factors of 88–135. This failure is the reason Layer 3 (the exact path integral) is essential.

11 Layer 3: VPM is the Exact Path Integral

This is the central theoretical result of the paper.

Evaluate the path integral over ϕ fluctuations around the saddle-point configured by a two-body DM density $n_\chi(r)$. The result is a one-loop effective action:

$$Z_{\text{scatter}} = e^{iS_{cl}} \cdot [\det(-\partial^2 - m_\phi^2 - y^2 n_\chi(r))]^{-1/2} \quad (19)$$

The determinant looks daunting, but it evaluates via the trace-log identity:

$$\ln \det = \text{Tr} \ln = \sum_\ell (2\ell + 1) \ln e^{2i\delta_\ell} = 2i \sum_\ell (2\ell + 1) \delta_\ell \quad (20)$$

The δ_ℓ here are exactly the VPM phase shifts from Chapter 1. The cross section follows immediately:

$$\boxed{\sigma_T = \frac{2\pi}{k^2} \sum_\ell w_\ell (2\ell + 1) \sin^2 \delta_\ell \equiv \text{exact one-loop path integral}} \quad (21)$$

There is no approximation, no truncation, no resummation. The VPM ODE of Chapter 1 is simply the most efficient way to evaluate a path-integral determinant. This identity will be central to future extensions of the model.

12 Layer 4: Does the Vacuum Prefer CP Violation?

Integrating out the Majorana fermion at one loop generates a Coleman-Weinberg potential for the CP angle θ . The calculation uses two fermionic degrees of freedom (Majorana):

$$V_{CW}(\theta) = -\frac{2}{64\pi^2} M_{eff}^4(\theta) \left[\ln \frac{M_{eff}^2(\theta)}{\mu^2} - \frac{3}{2} \right] \quad (22)$$

where, with A_4 flavor symmetry, $M_{eff}(\theta) = m_\chi \sqrt{\cos^2 \theta + \sin^2 \theta/9}$.

The key result: V_{CW} minimizes at $\theta = \pi/2$ — the vacuum *prefers* pure pseudoscalar coupling, i.e., maximal CP violation. The A_4 symmetry then discretizes this to $\theta = \arcsin(1/3)$. This is not a free parameter but an output of the one-loop calculation.

13 Layer 5: Is the Vacuum Stable at Freeze-out?

At high temperatures, thermal corrections can overwhelm the CW potential and shift the vacuum. We need to verify that θ_{relic} is stable during freeze-out at $T \sim m_\chi/25$.

The finite- T effective potential (Matsubara formalism):

$$V_{eff}(\theta, T) = V_{CW}(\theta) + \frac{T^4}{2\pi^2} \int_0^\infty dp p^2 [\ln(1 - e^{-E_\phi/T}) - 2 \ln(1 + e^{-E_\chi/T})] \quad (23)$$

At freeze-out temperature ($T = T_{fo}$), the thermal correction is $V_T/V_{CW} \sim 10^{-4}$: the CW minimum governs. The vacuum is stable. Only far above $T \gg m_\chi$ (before freeze-out) do thermal corrections push $\theta \rightarrow 0$ — but by then the field has not yet settled. There is no thermal attractor at $\theta \sim 2$ rad (verified numerically).

14 Layer 6: Can the Vacuum Tunnel Away?

Could the A_4 vacuum decay to a different state via quantum tunneling — an instanton — before the universe reaches its current age?

For a dark axion with $m_\sigma \sim H_0$ and $f \sim 0.24 M_{Pl}$, the Euclidean action is:

$$S_E \sim \left(\frac{f}{m_\sigma} \right)^2 \sim 10^{121} \quad (24)$$

The tunneling rate $\Gamma \sim e^{-S_E} = e^{-10^{121}}$ is not merely small — it is zero to any precision that could ever be measured. The vacuum is effectively eternal. This justifies treating θ_{relic} as static throughout cosmic history.

15 Layer 7: Relic Density as a Path-Integral Output

The s-wave annihilation amplitude, evaluated directly from the path integral diagram with two vertices and a t-channel propagator, gives:

$$\langle \sigma v \rangle_s = \frac{\pi \alpha^2}{4m_\chi^2} \quad (25)$$

This feeds into the Boltzmann equation for the co-moving yield $Y = n/s$. The solution gives $\Omega h^2 = 0.120$ at (m_χ, α_{relic}) for each point in the viable island, meaning the correct relic density is automatically achieved by the same α that governs SIDM.

16 The Hubble Constant from Dark QCD

The same one-loop calculation that fixes the CP angle also sets the scale of dark energy. The chain is:

$$\mathcal{L}_{SIDM} \xrightarrow{1\text{-loop}} V_{CW}(\theta) \xrightarrow{A_4} \theta_{relic} \xrightarrow{\text{dark QCD}} \Omega_\sigma = 0.69 \xrightarrow{\text{Friedmann}} H_0 = 67.4$$

The dark QCD sector confines at scale $\Lambda_d \sim 2.05$ meV, giving a dark axion mass $m_\sigma = \Lambda_d^2/f \sim H_0$ via the Gell-Mann–Oakes–Renner relation. The misalignment mechanism then gives:

$$\Omega_\sigma = \frac{f^2 \theta_i^2 H_0^2}{2\rho_{crit}} \approx 0.69 \quad \text{for} \quad \theta_i \approx 2 \text{ rad}, \quad f \approx 0.24 M_{\text{Pl}} \quad (26)$$

Substituting $\Omega_\sigma = \Omega_{DE} = 0.69$ into the Friedmann equation with $\Omega_m = 0.31$ gives $H_0 = 67.4$ km/s/Mpc directly. This is not a fit — f is the single free parameter, set by the requirement $V_\sigma \sim \rho_\Lambda$, and θ_i is the CW minimum.

Part III

Dark Electromagnetic Duality: Dark Energy as T-Breaking

17 The Central Analogy

In classical electromagnetism, a static charge produces only an electric field $\vec{E}$. Motion reveals a second component — the magnetic field $\vec{B}$ — which is not a new entity but a different projection of the same tensor $F_{\mu\nu}$. The critical property: $\vec{B}$ is T-odd while $\vec{E}$ is T-even.

We propose the dark sector obeys the same structure:

Table 5: Dark Electromagnetic Duality correspondence.

Electromagnetism	Dark Sector
$\vec{E}$ (T-even)	$y_s = y \cos(\sigma/f)$ — attraction, SIDM clustering
$\vec{B}$ (T-odd)	$y_p = y \sin(\sigma/f)$ — repulsion, DE acceleration
$F_{\mu\nu}$	$\mathcal{Y} = y_s + iy_p = ye^{i\gamma^5 \sigma/f}$
Lorentz boost mixes $E \leftrightarrow B$	σ field rotates $y_s \leftrightarrow y_p$
$\vec{B}$ does no work ($\vec{F} \perp \vec{v}$)	γ^5 prevents y_p from contributing to mass
Like charges repel electrically	y_s causes DM–DM attraction (SIDM)
Parallel currents attract magnetically	y_p causes vacuum repulsion (DE)

Thesis: Dark energy is the T-violating component of the dark matter interaction, made manifest through CP violation in the Majorana sector — exactly as magnetism is the T-violating component of electrostatics, made manifest through relative motion.

18 The Extended Lagrangian

In Chapter 1 the CP phase was a fixed parameter. Here we make it dynamical: what was a single number θ becomes a propagating field $\sigma(x)$, a dark pseudo-Goldstone boson with decay constant f . The physical consequence is that dark energy is not a cosmological constant bolted on by hand, but an oscillation of the CP angle around its equilibrium value.

The minimal modification to the Yukawa coupling is:

$$y_s + iy_p \longrightarrow ye^{i\gamma^5\sigma/f} = y \cos\left(\frac{\sigma}{f}\right) + iy \sin\left(\frac{\sigma}{f}\right) \gamma^5 \quad (27)$$

This is the unique $U(1)_A$ -invariant coupling consistent with the Majorana field equation. No mass term for σ is generated by SIDM interactions; the mass arises from dark QCD instanton effects at scale Λ_d (Chapter 2).

The full unified Lagrangian:

$$\mathcal{L} = \underbrace{\frac{1}{2}\bar{\chi}(i\not{\partial} - m_\chi)\chi + \frac{1}{2}(\partial\phi)^2 - V_0(\phi)}_{\text{SIDM (Ch.1)}} - \frac{1}{2}\bar{\chi}\left[y \cos\frac{\sigma}{f} + iy \sin\frac{\sigma}{f}\gamma^5\right]\chi\phi + \underbrace{\frac{1}{2}(\partial\sigma)^2 - V_{\text{bare}}(\sigma)}_{\text{dark axion}} \quad (28)$$

This single Lagrangian contains all three phenomena. The y_s -projection drives SIDM clustering (Chapter 1); the path-integral evaluation gives the T-even self-energy (Chapter 2); the y_p -projection, oscillating around θ_{relic} , supplies the accelerating vacuum energy (this chapter).

Parameter count: y fixed by SIDM+relic (Chapter 1); $f \sim 0.2 M_{\text{Pl}}$ from $V_\sigma \sim \rho_\Lambda$ (§20); all other free parameters already determined.

19 The Universal Relic Angle

Two constraints. One equation. No freedom.

The Chapter 1 MCMC enforces two independent conditions on the effective couplings: (i) the SIDM cross section must match $\sigma_T/m_\chi = 1\text{--}3 \text{ cm}^2/\text{g}$ across all benchmark halos, and (ii) the relic density must satisfy $\Omega h^2 = 0.120$. Expressing these as constraints on (y_s, y_p) :

$$\text{SIDM (Ch.1): } \alpha_s = \alpha \quad (29)$$

$$\text{Relic (Ch.1): } \alpha_s \alpha_p = \frac{\alpha^2}{8} \quad (30)$$

Dividing: $\alpha_p/\alpha_s = 1/8$. The relic angle is then fixed:

$$\theta_{\text{relic}} = \arctan \sqrt{\frac{\alpha_p}{\alpha_s}} = \arctan \frac{1}{\sqrt{8}} = 19.47^\circ \quad (31)$$

This is not a tuning. It is the *unique* equilibrium where the same coupling simultaneously satisfies DM self-interactions and the observed relic abundance. The value 19.47° is identical for every benchmark point in the MCMC island — it is as universal as the Weinberg angle, but derived, not measured.

20 Scale of the Dark Energy Potential

We now ask: given that σ is a dynamical field oscillating around θ_{relic} , what sets the amplitude of the potential energy? The answer comes from demanding that the vacuum energy density at $\theta = \theta_{\text{relic}}$ equals the observed dark energy density $\rho_\Lambda \simeq 2.9 \times 10^{-47} \text{ GeV}^4$.

The effective potential at the relic angle:

$$V_\sigma \approx V_{CW}(\theta_{\text{relic}}) \cdot \left(\frac{f}{M_{\text{Pl}}} \right)^4 \quad (32)$$

$$V_\sigma \approx \rho_\Lambda \quad \Leftrightarrow \quad f \approx 0.2 M_{\text{Pl}} \quad (33)$$

Numerically, for the MAP benchmark:

$$f_{\text{MAP}} = \frac{M_{\text{Pl}}}{(V_{CW}/\rho_\Lambda)^{1/4}} \approx 6.6 \times 10^{17} \text{ GeV} = 0.27 M_{\text{Pl}} \quad (34)$$

The sub-Planckian value $f \sim 0.2\text{--}0.3 M_{\text{Pl}}$ is natural in string compactifications and aligns with the weak gravity conjecture. This is therefore the *only* free parameter of the dark energy sector, and it is fixed by one observable: ρ_Λ .

21 Fifth Force and Screening

The σ field mediates a fifth force with coupling:

$$\beta = \frac{M_{\text{Pl}}}{f} \approx 5.0 \quad (\text{universal for all BPs}) \quad (35)$$

Existing Planck+BAO constraints give $\beta < 0.066$ for an unscreened coupled quintessence field. Resolution requires: (a) Chameleon screening in dense environments, or (b) σ coupling only to dark matter (secluded), not to baryons — consistent with the secluded dark sector hypothesis.

Since ϕ has no tree-level SM coupling (§2), σ — being a phase of the χ - ϕ system — naturally inherits this seclusion. Baryonic fifth-force constraints do not directly apply.

22 Observational Signatures

1. **Equation of state:** The dark axion potential predicts $w_{\text{DE}} = -1 + \mathcal{O}(\theta_i^2/3)$ near $\theta_i \sim 2$ rad. Current DESI constraint ($w_0 = -0.827 \pm 0.063$) is consistent at $\sim 1.6\sigma$ from the model prediction ($w_0 \approx -0.727$).
2. **Hubble tension:** $H_0 = 67.4 \text{ km/s/Mpc}$ (from Chapter 2) is consistent with CMB measurements and does not worsen the H_0 tension relative to ΛCDM .
3. **CP violation in SIDM:** The y_s/y_p ratio is in principle measurable through indirect detection asymmetries at $\mathcal{O}(\sin^2 \theta_{\text{relic}}) \approx 10\%$ level.
4. **Null prediction:** No baryonic fifth force. No cross-coupling between σ and SM particles at tree level.

23 Summary: From SIDM to Dark Energy in One Lagrangian

The three parts of this paper are not separate models. They are three projections of one Lagrangian:

- **T-even projection** (y_s, ϕ): DM self-interactions, velocity-dependent cross sections, Fornax cores, relic density. This is Chapter 1.
- **Path-integral structure**: VPM = fluctuation determinant, CW vacuum, Matsubara freeze-out, dark QCD $\rightarrow H_0$. This is Chapter 2.
- **T-odd projection** (y_p, σ): vacuum energy, DE equation of state, universal relic angle. This is Chapter 3.

The unification is not formal — the same coupling y , the same mass m_χ , and the same mediator mass m_ϕ that explain the rotation curves of SPARC galaxies and the survival of Fornax globular clusters also set the scale of dark energy and predict $H_0 = 67.4$ km/s/Mpc.

Discussion

A Comparison with Alternative Dark Matter Models

Table 6 summarizes how our framework compares with five leading alternatives across the key phenomenological tests.

Table 6: Model comparison across key observables. $\checkmark$ = naturally explained, $\sim$ = requires tuning, $\times$ = fails or does not address, $—$ = not applicable. Only the present model simultaneously addresses SIDM small-scale problems, relic density, direct detection evasion, and dark energy within a single Lagrangian.

Observable	CDM	WIMP	FDM	aSIDM	vdSIDM	Ours
Core-cusp	$\times$	$\times$	$\sim$	$\checkmark$	$\checkmark$	$\checkmark$
Missing satellites	$\sim$	$\sim$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$
Too-big-to-fail	$\times$	$\times$	$\sim$	$\checkmark$	$\checkmark$	$\checkmark$
Halo diversity	$\times$	$\times$	$\times$	$\sim$	$\checkmark$	$\checkmark$
Cluster mergers	$\checkmark$	$\checkmark$	$\checkmark$	$\sim$	$\checkmark$	$\checkmark$
Fornax GC survival	$\times$	$\times$	$\sim$	$\sim$	$\checkmark$	$\checkmark$
Correct relic Ωh^2	postulated	$\checkmark$	postulated	$\sim$	$\sim$	$\checkmark$
Direct detection safe	$—$	$\sim$	$—$	$\sim$	$\sim$	$\checkmark$
ΔN_{eff} safe	$\checkmark$	$\checkmark$	$\checkmark$	$\sim$	$\sim$	$\checkmark$
Dark energy	separate	separate	separate	separate	separate	$\checkmark$
H_0 prediction	input	input	input	input	input	67.4
# free parameters	6	≥ 4	1	3	3	3

We briefly discuss each alternative:

Λ CDM. Remarkably successful on large scales (> 1 Mpc) but fails on small scales — core-cusp, too-big-to-fail, and diversity problems persist even with baryonic feedback. The cosmological constant is an input, not a prediction. Our model reduces to Λ CDM phenomenology on large scales (velocity-suppressed σ_T) while resolving the small-scale tensions.

WIMPs (neutralino, Higgsino, etc.). Provide the correct relic density via the “WIMP miracle,” but have no self-interaction mechanism and thus inherit all CDM small-scale problems. Furthermore, they face growing tension with LZ/XENONnT direct detection null results (except for finely tuned blind spots). Our model achieves the relic density without any direct detection signal by construction.

Fuzzy dark matter (FDM, $m \sim 10^{-22}$ eV). Creates cores via quantum pressure (de Broglie wavelength $\sim$ kpc). However, Lyman- α forest observations constrain $m > 2 \times 10^{-20}$ eV [7], in strong tension with the core sizes needed for dwarf galaxies. FDM also cannot explain halo diversity (all halos get the same core for given mass) and does not address dark energy.

Constant SIDM (aSIDM, $\sigma/m = \text{const}$). Solves dwarf-scale problems but is excluded by cluster mergers if $\sigma/m \gtrsim 1$ cm²/g. The velocity dependence in our model is automatic — it emerges from the Yukawa potential, not from an ad hoc form factor.

Velocity-dependent SIDM (vdSIDM), generic. This is the closest competitor. Many groups have explored Yukawa-mediated SIDM, typically with Dirac fermions [2, 4]. Our model differs in three respects: (i) the fermion is Majorana, producing a distinct spin-statistics fingerprint (§7.7); (ii) the dark sector is fully secluded, eliminating direct detection and CMB constraints; (iii) the dark energy sector emerges from CP violation in the same Lagrangian rather than being a separate model.

B The A_4 Symmetry and the Neutrino Question

The A_4 discrete symmetry plays a central role in our framework: it governs the Coleman-Weinberg vacuum structure (§12) and the CP angle θ_{relic} .

The same A_4 group has been extensively studied in neutrino physics as the origin of tribimaximal mixing [8, 9]. In those models, A_4 acts on three lepton generations and predicts specific neutrino mixing angles. In our framework, A_4 acts on the *dark sector* — the dark fermion χ and mediator ϕ . The coincidence is suggestive but, at present, coincidental.

We emphasize what this model **does not** contain:

- No neutrino mass mechanism. The secluded dark sector has zero tree-level coupling to the Standard Model, so no see-saw or radiative neutrino mass is generated.
- No lepton number violation visible to SM. The Majorana mass of χ is in the dark sector; χ carries no SM quantum numbers.
- No θ_{13} correction to dark CP. The neutrino mixing angle θ_{13} does not feed back into the dark sector at tree level.

However, the fact that A_4 governs *both* neutrino mixing (in the SM sector) and dark CP violation (in our dark sector) raises the possibility of a unified A_4 UV completion where a single symmetry breaking simultaneously sets the neutrino mass matrix and the dark sector CW vacuum. This is left as an explicit open question for future work. The

key observable would be a correlation between the neutrino CP phase δ_{CP} and the dark sector relic angle $\theta_{relic} = 19.47^\circ$ — testable with DUNE and Hyper-K.

C Relation to Dark QCD and Confinement

The dark energy prediction (Chapter 2, §16) relies on a dark QCD sector that confines at $\Lambda_d \sim 2.05$ meV. The dark axion σ is the pseudo-Goldstone boson of the spontaneously broken dark chiral symmetry, with mass $m_\sigma = \Lambda_d^2/f \sim H_0$ via the Gell-Mann–Oakes–Renner relation.

This sector is not added by hand — it is the non-perturbative completion of the A_4 -symmetric dark Yukawa interaction. The QCD-like dynamics provides: (i) confinement of dark color (eliminating free dark quarks as a relic); (ii) the instanton-generated potential for σ ; (iii) the mass gap that protects σ from radiative corrections.

The coincidence $\Lambda_d \sim \text{meV} \sim H_0$ appears fine-tuned but is in fact a consequence of the Friedmann equation: $V_\sigma = f^2 m_\sigma^2/2 = \rho_\Lambda$ determines m_σ once $f \sim 0.2 M_{\text{Pl}}$ is set by Planck-scale dynamics. It is no more fine-tuned than the electroweak scale in the Standard Model — both are set by symmetry breaking scales.

D Limitations and Open Problems

We list the principal limitations of the present work and their status:

1. **Mediator overclosure (FATAL).** The ϕ relic abundance must be depleted by cannibal processes $3\phi \leftrightarrow 2\phi$. A full thermal history calculation (beyond the simple estimate $\Omega_\phi/\Omega_\chi \sim 5 \times 10^{-4}$) has not been performed. If the cannibal cross section is too small, ϕ could overclose the universe. This requires a dedicated Boltzmann solver tracking both χ and ϕ simultaneously.
2. **Mixed Majorana coupling.** The Chapter 3 decomposition assumes a pure scalar coupling y_s for SIDM at $\theta = 0$ and pure pseudoscalar y_p at $\theta = \pi/2$. In reality, the relic angle $\theta_{relic} = 19.47^\circ$ means the physical coupling is a mixture: $y_{\text{phys}} = y \cos \theta_{relic} + iy \sin \theta_{relic} \gamma^5$. The interference between scalar and pseudoscalar exchange modifies the self-interaction cross section at the $\sim 10\%$ level. A full calculation of σ_T with the mixed $(y_s + iy_p \gamma^5)$ vertex is needed.
3. **Spiral galaxy fits.** The RAR analysis uses a simple SIDM core model without adiabatic contraction (Gnedin et al. 2004). For spiral galaxies like NGC 2403 and NGC 3198, the baryonic gravitational potential modifies the DM density profile significantly. A self-consistent calculation with adiabatic contraction could shift σ_T/m by up to $\sim 30\%$.
4. **DESI DR2 equation of state.** The model predicts $w_0 \approx -0.727$, while DESI DR2 data (2025) favor $w_0 = -0.827 \pm 0.063$, i.e., a $\sim 1.6\sigma$ tension. A steeper thawing potential (or a more complex dark axion potential beyond the cosine approximation) may be needed if this tension persists. This is a *testable prediction*: future DESI data releases will either confirm or rule out the predicted w_0 .
5. **MCMC convergence at high λ .** The MCMC sampler explores λ up to 49 (MAP point), where the VPM solver requires $\ell_{\text{max}} \sim 80$ partial waves. The convergence

diagnostics ($\hat{R} < 1.01$, $\tau_{\max} = 75$) are satisfactory, but a production run with > 3000 steps and additional QA (one-sided χ^2 treatment, $\lambda > 50$ cutoff) is desirable.

6. **No UV completion.** The model is an effective field theory. The Yukawa coupling, A_4 symmetry, and dark QCD confinement are postulated at the Lagrangian level. A UV completion (e.g., embedding in a string compactification or a composite Higgs-like framework) is not provided.

None of these limitations invalidate the existing results; all cross sections and relic density calculations are exact within the stated model. The limitations define the boundary of the present analysis and the agenda for subsequent work.

E Conclusions

We have presented a unified framework for the dark sector built from a single Lagrangian with three free parameters (m_χ, m_ϕ, α):

1. A Majorana fermion χ interacting via a secluded scalar mediator ϕ provides velocity-dependent self-interactions that resolve the core-cusp, too-big-to-fail, and halo diversity problems (Chapter 1).
2. The same Lagrangian, evaluated as a path integral, proves that the VPM scattering ODE is the exact fluctuation determinant (no approximation), that the CW potential selects a CP-violating vacuum, and that dark QCD confinement predicts $H_0 = 67.4$ km/s/Mpc (Chapter 2).
3. The CP phase, promoted to a dynamical dark axion σ , provides an “electromagnetic duality” of the dark sector: the T-even component y_s drives SIDM clustering while the T-odd component y_p sources the observed vacuum energy (Chapter 3).

The model is tested against 13 astrophysical observables spanning $v = 12\text{--}4700$ km/s, achieving $\chi_{\min}^2/\nu = 0.12$ (unconstrained) and $\chi_{\text{relic}}^2/\nu = 0.38$ (relic-constrained). All 17 relic-viable benchmark points satisfy every observational constraint simultaneously. The numerical pipeline is validated by a six-part cross-check suite including Born-limit agreement, unitarity, transfer $\equiv$ elastic identity, literature comparison, and blind stress tests.

The framework makes falsifiable predictions: a Majorana spin-statistics fingerprint at $v \sim 40\text{--}95$ km/s (observable in intermediate dSphs), no SMBH seed formation ($t_{\text{gc}}/t_{\text{univ}} > 10^4$), $w_0 \approx -0.73$ (testable with DESI), and a universal relic angle $\theta_{\text{relic}} = 19.47^\circ$ independent of (m_χ, m_ϕ, α) .

All code, data, and cross-check scripts are publicly available [10].

F VPM as Fluctuation Determinant: Proof

The key identity proved in Chapter 2 (PI-6):

$$\ln \det [-\partial^2 - m_\phi^2 - y^2 n_\chi(r)]^{-1/2} = i \sum_\ell (2\ell + 1) \delta_\ell \quad (36)$$

where δ_ℓ are the VPM phase shifts. This establishes that the Variable Phase Method is not an approximation but the *exact* evaluation of the one-loop path integral over the mediator field ϕ in the presence of the two-body DM density $n_\chi(r)$.

G Universal Relic Angle: Derivation

From the two constraints:

$$\text{SIDM: } \alpha_s = \alpha \quad (37)$$

$$\text{Relic: } \alpha_s \alpha_p = \alpha^2/8 \quad (38)$$

it follows immediately that $\alpha_p = \alpha/8$, hence:

$$\theta_{\text{relic}} = \arctan \sqrt{\alpha_p/\alpha_s} = \arctan(1/\sqrt{8}) = 19.47^\circ \quad (39)$$

This angle is *universal* — independent of (m_χ, m_ϕ, α) .

H Instanton Suppression of θ_{A_4}

For a dark axion with $m_\sigma \sim H_0$ and $f \sim 0.24 M_{\text{Pl}}$:

$$S_E \sim \left(\frac{f}{m_\sigma} \right)^2 \sim 10^{121} \quad \Rightarrow \quad \Gamma_{\text{tunnel}} \sim e^{-S_E} \approx 0 \quad (40)$$

The A_4 vacuum is stable to tunneling on timescales $10^{121} \times t_{\text{Hubble}}$ — effectively eternal.

I Full Benchmark Table

Table 7 lists the 17 relic-constrained benchmark points with $\Omega h^2 = 0.120$ and their cross sections at characteristic velocities.

J Maxwell-Boltzmann Velocity Averaging

Our fits use the fixed-velocity cross section $\sigma_T(v_0)$ rather than the Maxwell-Boltzmann average $\langle \sigma_T \rangle_{\text{MB}}$. Table 8 quantifies the systematic bias $\varepsilon(v_0) = \langle \sigma_T/m \rangle_{\text{MB}} / [\sigma_T(v_0)/m] - 1$ for BP1:

All 17 relic benchmarks remain at $\chi^2/\nu < 1$ under MB averaging. No viability status changes.

K Sommerfeld Enhancement Details

The non-relativistic Sommerfeld factor in the s -wave channel is:

$$S_0 = \frac{2\pi\alpha/v}{1 - e^{-2\pi\alpha/v}} \cdot F(\alpha, m_\phi, v) \quad (41)$$

where F accounts for the finite Yukawa range ($F \rightarrow 1$ as $m_\phi \rightarrow 0$).

At freeze-out ($v \sim 0.3c$):

Benchmark	κ_{fo}	S_0	$\Delta\Omega h^2/\Omega h^2$
BP1 ($m_\chi = 20.7$ GeV)	93	1.003	0.3%
BP9 ($m_\chi = 37.9$ GeV)	168	1.010	1.0%
MAP ($m_\chi = 94.1$ GeV)	760	1.025	2.5%

The tree-level Boltzmann calculation is valid for all benchmarks.

Table 7: 17 relic-viable benchmark points with cross sections σ_T/m in cm^2/g at $v = 30$ and 1000 km/s. All satisfy $\chi^2/\nu < 1$ and the Bullet Cluster upper limit.

m_χ [GeV]	m_ϕ [MeV]	α	λ	$\sigma_T/m(30)$	$\sigma_T/m(1000)$
10.0	7.56	5.54×10^{-4}	0.73	0.52	0.072
14.4	8.66	7.56×10^{-4}	1.26	0.73	0.080
20.7	9.91	1.05×10^{-3}	2.19	0.79	0.088
20.7	11.34	1.05×10^{-3}	1.91	0.52	0.072
29.8	11.34	1.47×10^{-3}	3.86	0.75	0.090
37.9	12.98	1.87×10^{-3}	5.46	0.61	0.095
42.8	13.15	2.09×10^{-3}	6.80	0.58	0.093
48.3	14.85	2.36×10^{-3}	7.67	0.50	0.089
54.5	13.75	2.63×10^{-3}	10.4	0.62	0.096
61.6	14.10	2.97×10^{-3}	13.0	0.58	0.095
69.5	13.62	3.35×10^{-3}	17.1	0.64	0.097
78.5	12.50	3.78×10^{-3}	23.7	0.71	0.098
88.5	11.10	4.26×10^{-3}	32.4	0.70	0.098
94.1	11.10	5.73×10^{-3}	48.6	—	—
100.0	9.15	2.86×10^{-3}	31.2	1.33	0.153
100.0	10.0	3.50×10^{-3}	35.0	—	—
100.0	12.0	4.81×10^{-3}	40.1	—	—

Table 8: Velocity-averaging correction ε for BP1. The fixed-velocity approximation introduces $< 6\%$ bias at dwarf velocities and $< 15\%$ at cluster scales — small compared to observational uncertainties of $\sim 0.3\text{--}0.5$ dex.

System	v [km/s]	Fixed σ_T/m	MB-avg $\langle\sigma_T/m\rangle$	ε
Draco	12	0.430	0.449	+4%
TBTF	30	0.516	0.506	−2%
NGC 2976	60	0.499	0.471	−6%
NGC 720	250	0.325	0.282	−13%
Abell 2537	1100	0.061	0.052	−15%
Bullet	4700	0.004	0.004	−2%

L Numerical Error Budget

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Table 9: Systematic error budget for σ_T/m at $v = 30$ km/s.

Source	Relative error	Direction
$\ell_{\max}$ truncation	$< 0.01\%$	either
Integration step (h)	$< 0.1\%$	either
$x_{\max}$ boundary	$< 10^{-6}$	positive
Born vs VPM (not an error)	200–400%	overestimate
MB averaging vs fixed- v	2–6%	underestimate
Boltzmann (KT vs numerical)	16% in α	systematic
Total numerical	$< 0.2\%$	
Total physical	$\sim 16\%$	(Boltzmann)

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